

Lab 3: Bash Scripting - Build a Reconnaissance Tool

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Lab Title: Lab 3 - Bash Scripting: Build a Reconnaissance Tool

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Authorized Target: 10.15.203.91

Executive Summary

In this lab, I developed a Bash script that automates three common reconnaissance tools, namely **WhatWeb**, **Nmap**, and **DIRB**, into a single menu-driven application. Instead of running each command manually, the script prompts the user to enter an authorized target, validates the input to ensure it is not empty, displays a menu of available reconnaissance tools, and executes the selected tool against the specified target.

The script was developed incrementally rather than all at once. I began by creating the header and banner message, followed by implementing user input for the target address. I then added input validation to prevent empty submissions, created a menu interface, and finally implemented a case statement to handle the execution of the selected reconnaissance tool. Once the script was completed, it was saved, made executable using `chmod +x`, and executed directly with `./recon_tool.sh`.

The completed script was tested against the authorized lab target **10.15.203.91**. Testing confirmed that all three menu options functioned correctly. **WhatWeb** successfully identified web server technologies, **Nmap** performed service detection and revealed multiple open ports and running services, and **DIRB** successfully discovered directories and resources hosted on the target system. This lab strengthened my understanding of fundamental Bash scripting concepts, including variables, user input, conditional statements (if), case statements, and functions. It also reinforced the importance of ensuring that reconnaissance activities are performed only against systems for which explicit authorization has been granted.

Objective

The objective of this lab was to create a Bash script that:

- Prompts the user to enter a target IP address or domain.
- Ensures the target is not hardcoded within the script.
- Validates that a target has been supplied.
- Displays a menu containing:

- WhatWeb
 - Nmap
 - DIRB
 - Exit
- Uses a case statement to handle menu selections.
 - Verifies that required tools are installed before execution.
 - Runs the selected reconnaissance tool against the user-supplied target.
 - Is executable and can be run directly from the terminal.

Setting Up the Environment

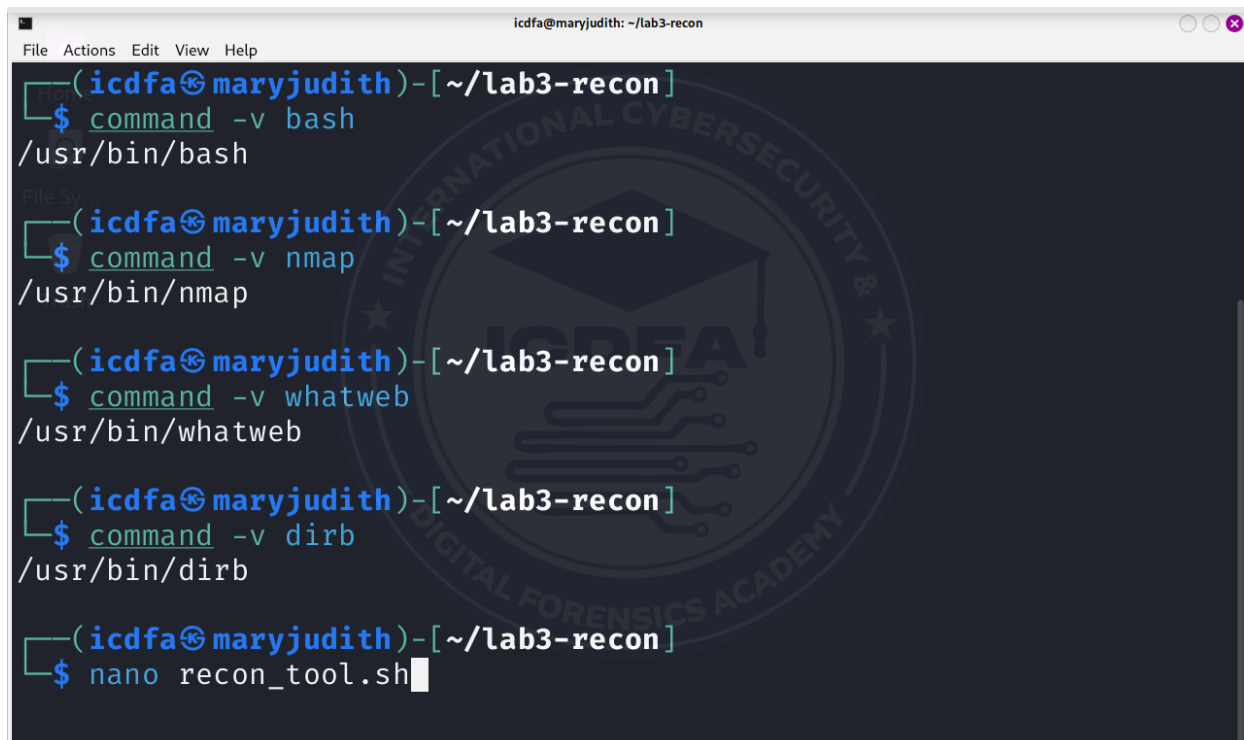
Before writing the script, I created a dedicated working directory and verified that all required tools were installed.

```
mkdir -p ~/lab3-recon      cd ~/lab3-recon
```

I then checked the availability of the required tools:

```
command -v bash      command -v nmap      command -v whatweb      command -v dirb
```

The command -v command displays the location of a program if it is installed. All four commands returned valid paths, confirming that the required tools were available.

A terminal window titled 'icdfa@maryjudith: ~/lab3-recon' with a menu bar (File, Actions, Edit, View, Help). The terminal shows a series of commands and their outputs. A large, faint watermark of the ICDEFA logo is visible in the background. The commands and outputs are: 1. '\$ command -v bash' followed by '/usr/bin/bash'. 2. '\$ command -v nmap' followed by '/usr/bin/nmap'. 3. '\$ command -v whatweb' followed by '/usr/bin/whatweb'. 4. '\$ command -v dirb' followed by '/usr/bin/dirb'. 5. '\$ nano recon_tool.sh' with a cursor at the end of the line.

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
(icdfa@maryjudith)-[~/lab3-recon]
$ command -v bash
/usr/bin/bash
(icdfa@maryjudith)-[~/lab3-recon]
$ command -v nmap
/usr/bin/nmap
(icdfa@maryjudith)-[~/lab3-recon]
$ command -v whatweb
/usr/bin/whatweb
(icdfa@maryjudith)-[~/lab3-recon]
$ command -v dirb
/usr/bin/dirb
(icdfa@maryjudith)-[~/lab3-recon]
$ nano recon_tool.sh
```

Figure 1 — Confirming bash, nmap, whatweb and dirb are all installed.

Building the Script, Stage by Stage

Rather than writing the whole script in one go, I built it up gradually in nano, testing at each stage before moving to the next.

Stage A – Header and banner

The script opens with a shebang line and three echo statements that print a title, a divider, and a reminder that it must only be used against authorised targets. The shebang (`#!/usr/bin/env bash`) tells Linux to run the file using Bash.

Stage B – Reading the target

`read -rp` prompts the user and stores their answer in the variable `target`. The `-r` stops backslashes from being treated specially, and `-p` just displays the prompt text on the same line.

Stage C – Validating the input

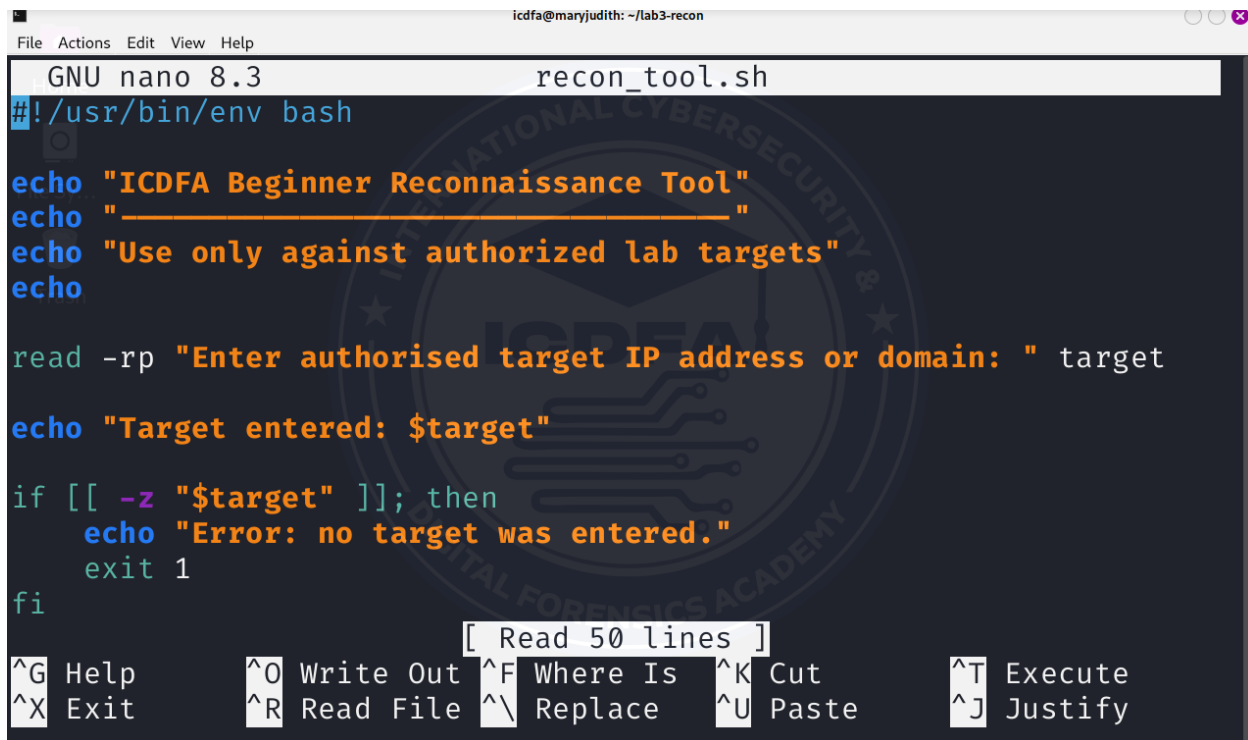
An if `[[ -z "$target" ]]` check tests whether the `target` variable is empty. If the user just presses Enter without typing anything, the script prints an error and exits with exit 1, so it never runs a scan with a blank target.

Stage D – Building the menu

A short block of echo statements lists four options — WhatWeb, Nmap, DIRB and Exit — and read -rp stores the user's number choice in a variable called choice.

Stage E & F – case logic and the tool-check function

A case "\$choice" in ... esac statement compares choice against 1, 2, 3 and 4, with a catch-all * branch for anything invalid. Before running any tool, a small check tool() function uses command -v to confirm the tool is actually installed — if it isn't, the script prints an error and stops instead of failing halfway through a scan.



```
GNU nano 8.3 recon_tool.sh
#!/usr/bin/env bash

echo "ICDFA Beginner Reconnaissance Tool"
echo "_____ "
echo "Use only against authorized lab targets"
echo

read -rp "Enter authorised target IP address or domain: " target

echo "Target entered: $target"

if [[ -z "$target" ]]; then
    echo "Error: no target was entered."
    exit 1
fi

[ Read 50 lines ]

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

Figure 2 — The header, target prompt and empty-input validation, as typed in nano.

Stage G – Putting it all together

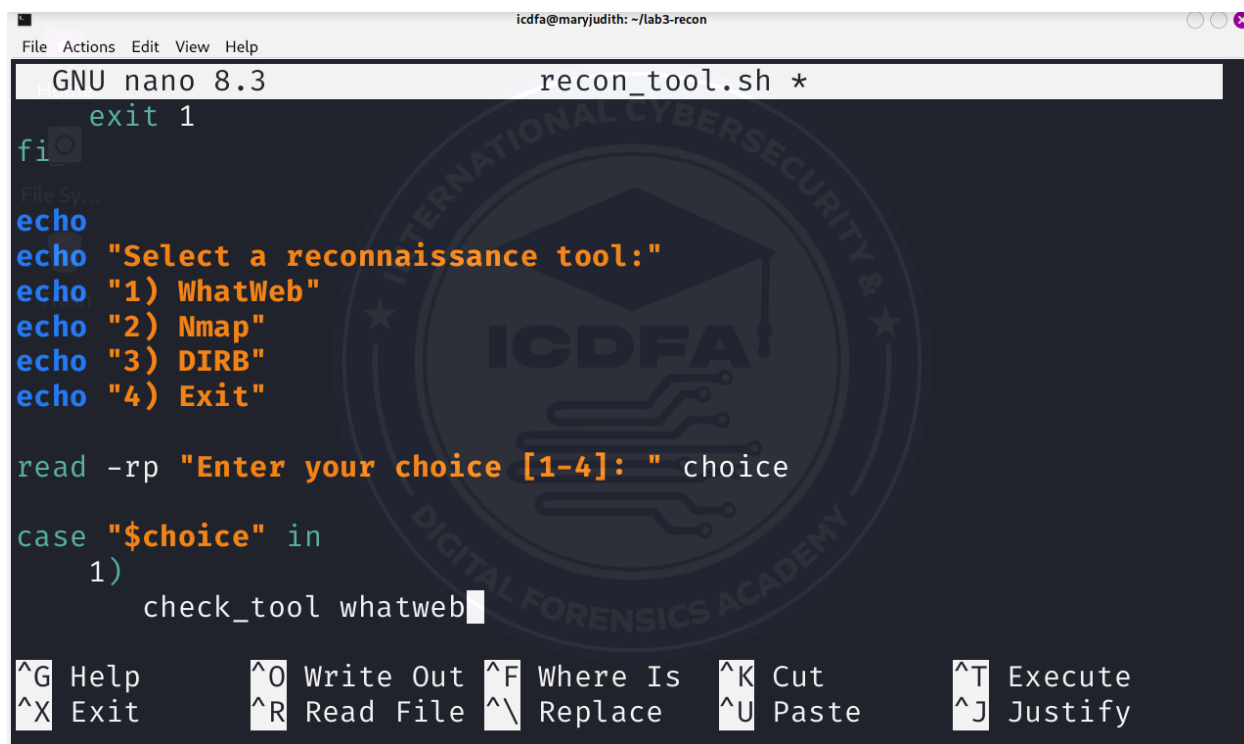
The final case block ties everything together: option 1 checks for and runs WhatWeb against http://\$target, option 2 checks for and runs nmap -sV \$target, option 3 checks for and runs DIRB against http://\$target, and option 4 exits cleanly without scanning anything. Any other input falls through to the * branch and is rejected as an invalid choice.

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
GNU nano 8.3 recon_tool.sh *
exit 1
fi
File Sy...
echo
echo "Select a reconnaissance tool:"
echo "1) WhatWeb"
echo "2) Nmap"
echo "3) DIRB"
echo "4) Exit"

read -rp "Enter your choice [1-4]: " choice

case "$choice" in
1)
    check_tool whatweb

```



```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
GNU nano 8.3 recon_tool.sh *
case "$choice" in
1)
    check_tool whatweb
    echo "[+] Running WhatWeb against $target"
    whatweb "http://$target"
    ;;
2)
    check_tool nmap
    echo "[+] Running Nmap service detection against $target"
    nmap -sV "$target"
    ;;
3)
    check_tool dirb
    echo "[+] Running DIRB against $target"

```

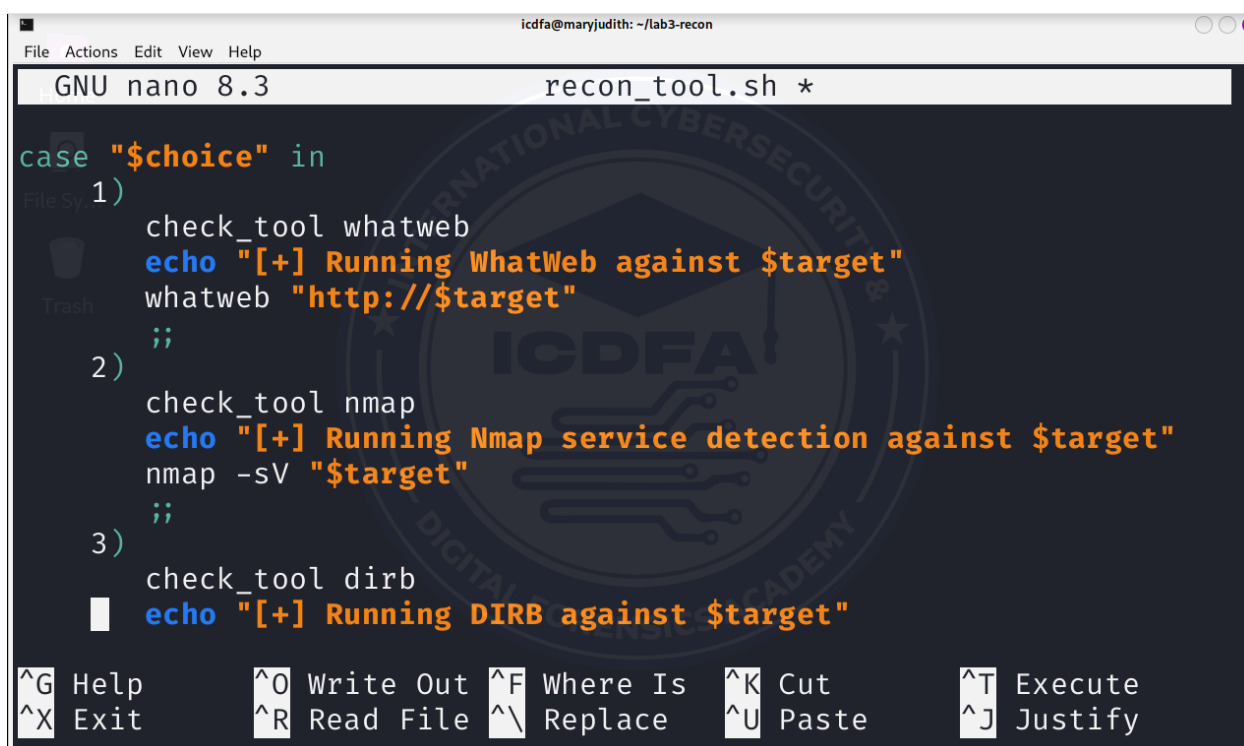
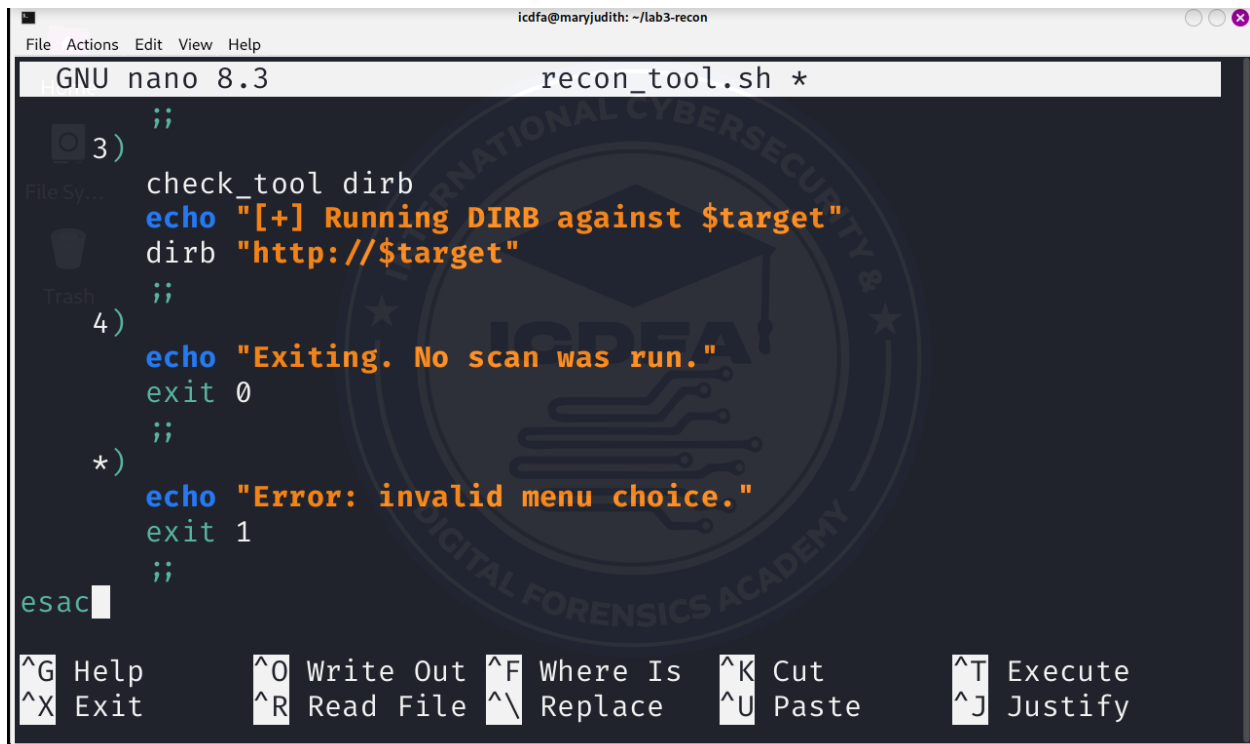


Figure 3 — The completed case statement covering WhatWeb, Nmap and DIRB.



```
icdfa@maryjudith: ~/lab3-recon
GNU nano 8.3 recon_tool.sh *
;;
3)
check_tool dirb
echo "[+] Running DIRB against $target"
dirb "http://$target"
;;
4)
echo "Exiting. No scan was run."
exit 0
;;
*)
echo "Error: invalid menu choice."
exit 1
;;
esac
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

Saving, Permissions and Running the Script After writing the script in nano,

I saved it with Ctrl+O, Enter, then exited with Ctrl+X. Next, I made the file executable and confirmed the permission had been applied:

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
/usr/bin/dirb

(icdfa@maryjudith)-[~/lab3-recon]
$ nano recon_tool.sh

(icdfa@maryjudith)-[~/lab3-recon]
$ chmod +x recon_tool.sh

(icdfa@maryjudith)-[~/lab3-recon]
$ ls l recon_tool.sh
ls: cannot access 'l': No such file or directory
recon_tool.sh

(icdfa@maryjudith)-[~/lab3-recon]
$ ls -l recon_tool.sh
-rwxrwxr-x 1 icdfa icdfa 1030 Sep  5 21:55 recon_tool.sh

(icdfa@maryjudith)-[~/lab3-recon]
$
```

Figure 4 — Running `chmod +x recon_tool.sh` and confirming the executable permission with `ls -l` (note: the permission string shows `-rwxrwxr-x`, meaning it now includes execute rights)

With execute permission in place, I ran the script directly from the terminal using `./recon_tool.sh` — the `./` tells the shell to run the file from the current folder rather than searching the system PATH for it.

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
   group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
   state UP group default qlen 1000
   link/ether 00:0c:29:cd:75:ad brd ff:ff:ff:ff:ff:ff
   inet 10.15.203.36/24 brd 10.15.203.255 scope global dynamic nop
   refixroute eth0
       valid_lft 3473sec preferred_lft 3473sec
   inet6 fe80::20c:29ff:fedc:75ad/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

(icdfa@maryjudith)-[~/lab3-recon]
$ ping -c 4 10.15.203.91
```

Figure 5 — Checking network details and confirming the target 10.15.203.91 is reachable.

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
refixroute eth0
   valid_lft 3473sec preferred_lft 3473sec
   inet6 fe80::20c:29ff:fedc:75ad/64 scope link noprefixroute
   valid_lft forever preferred_lft forever

(icdfa@maryjudith)-[~/lab3-recon]
$ ping -c 4 10.15.203.91
PING 10.15.203.91 (10.15.203.91) 56(84) bytes of data.
64 bytes from 10.15.203.91: icmp_seq=1 ttl=64 time=0.862 ms
64 bytes from 10.15.203.91: icmp_seq=2 ttl=64 time=1.72 ms
64 bytes from 10.15.203.91: icmp_seq=3 ttl=64 time=1.59 ms
64 bytes from 10.15.203.91: icmp_seq=4 ttl=64 time=1.97 ms

— 10.15.203.91 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 0.862/1.536/1.972/0.412 ms

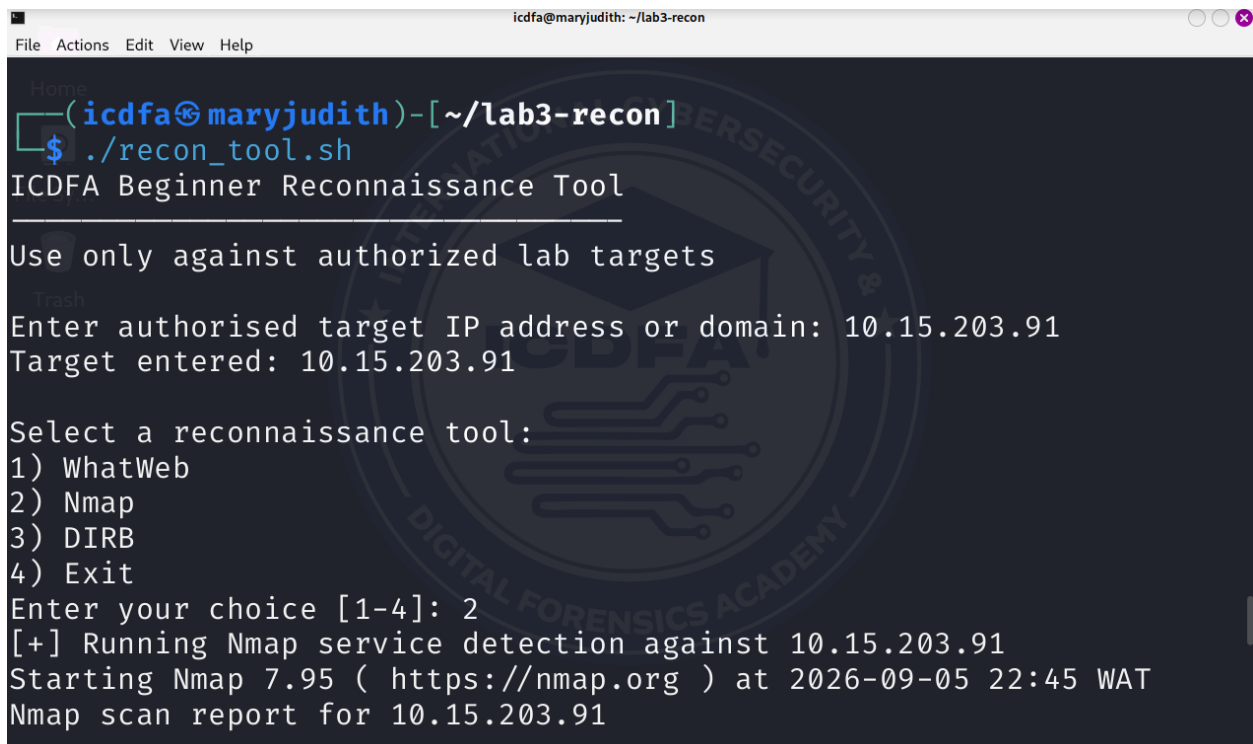
(icdfa@maryjudith)-[~/lab3-recon]
$
```

I also confirmed the lab machine could actually reach the target before scanning it, using a simple ping test 4 packets sent, 4 received, 0% loss.

Testing Against the Authorized Target

I tested the script against the authorized lab target, 10.15.203.91, and worked through all three menu options one at a time.

Test 2 : **Nmap (Option 2)** Selecting option 2 triggered nmap -sV against the target. The scan came back with a long list of open ports and their detected service versions including FTP (vsftpd 2.3.4), SSH (OpenSSH 4.7p1), Telnet, SMTP (Postfix), DNS (ISC BIND), HTTP (Apache 2.2.8) and several others. This confirms -sV is working correctly, since it's returning specific version numbers rather than just port states.



```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
Home
(icdfa@maryjudith)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool
Use only against authorized lab targets
Enter authorised target IP address or domain: 10.15.203.91
Target entered: 10.15.203.91
Select a reconnaissance tool:
1) WhatWeb
2) Nmap
3) DIRB
4) Exit
Enter your choice [1-4]: 2
[+] Running Nmap service detection against 10.15.203.91
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-05 22:45 WAT
Nmap scan report for 10.15.203.91
```

Figure 6 — Launching the script, entering the target, and selecting the Nmap option from the menu

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
Nmap scan report for 10.15.203.91
Host is up (0.00080s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
```

Figure 7 — Nmap service-detection results: open ports and their versions on the target.

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:49:C3:1A (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
2049/tcp open  nfs      2-4 (RPC #100003)
2121/tcp open  ftp      ProFTPD 1.3.1
3306/tcp open  mysql    MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc      VNC (protocol 3.3)
6000/tcp open  X11      (access denied)
6667/tcp open  irc      UnrealIRCd
8009/tcp open  ajp13    Apache Jserv (Protocol v1.3)
8180/tcp open  http     Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:49:C3:1A (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.68 seconds

(icdfa@maryjudith)-[~/lab3-recon]
$
```

(Option 2) WhatWeb ran successfully and fingerprinted the web server, identifying it as an Ubuntu Apache 2.2.8 server running PHP 5.2.4, with the page title "Metasploitable2 - Linux" confirming the target is the intentionally vulnerable Metasploitable2 lab machine.

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
Home
Trash
(icdfa@maryjudith)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool

Use only against authorized lab targets

Enter authorised target IP address or domain: 10.15.203.91
Target entered: 10.15.203.91

Select a reconnaissance tool:
1) WhatWeb
2) Nmap
3) DIRB
4) Exit
Enter your choice [1-4]: 1
[+] Running WhatWeb against 10.15.203.91
http://10.15.203.91 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ],
HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.15.203
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
Use only against authorized lab targets

Enter authorised target IP address or domain: 10.15.203.91
Target entered: 10.15.203.91

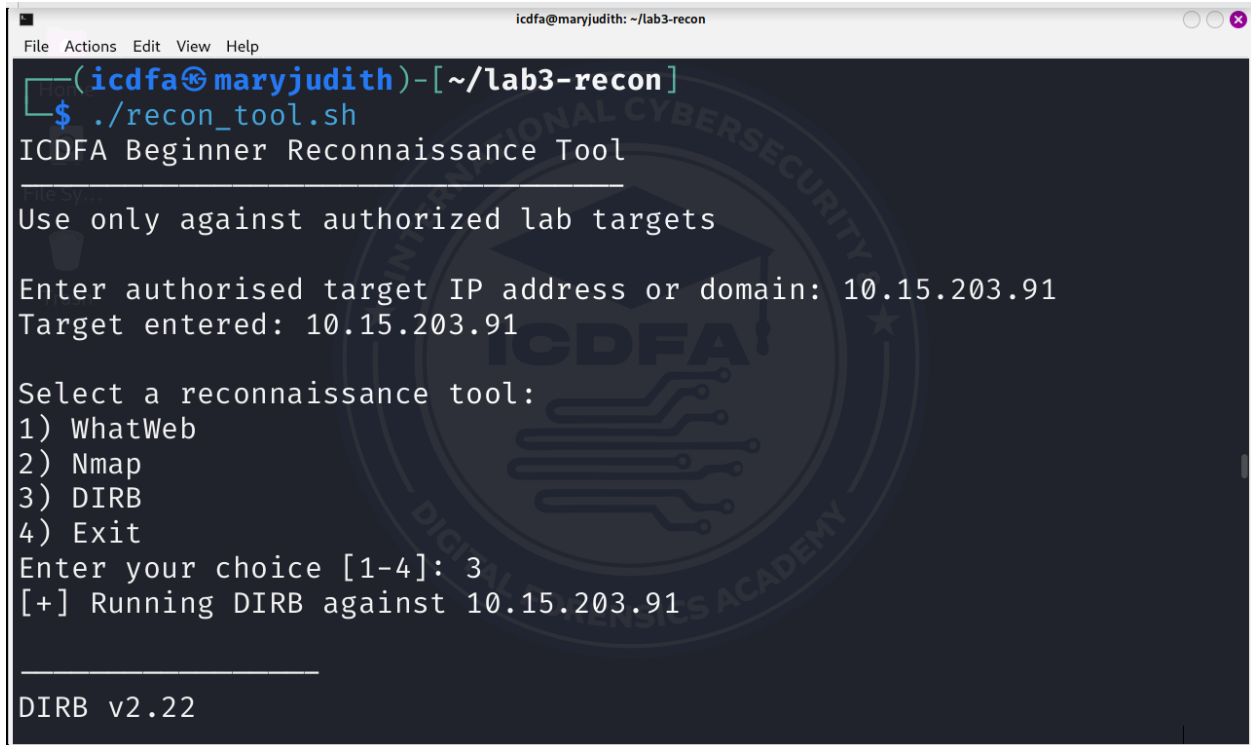
Select a reconnaissance tool:
1) WhatWeb
2) Nmap
3) DIRB
4) Exit
Enter your choice [1-4]: 1
[+] Running WhatWeb against 10.15.203.91
http://10.15.203.91 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ],
HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.15.203
.91], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDA
V[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

(icdfa@maryjudith)-[~/lab3-recon]
$
```

Figure 8 — Successful WhatWeb scan showing the server banner, PHP version and page title.

Test 3 — DIRB (Option 3)

Option 3 ran dirb against http://10.15.203.91. DIRB worked through its wordlist and successfully discovered real directories on the server, including a phpMyAdmin folder with several accessible files inside it (calendar, changelog, docs, index.php and more) solid evidence the tool executed correctly against a live target.



```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
(icdfa@maryjudith)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool
-----
Use only against authorized lab targets

Enter authorised target IP address or domain: 10.15.203.91
Target entered: 10.15.203.91

Select a reconnaissance tool:
1) WhatWeb
2) Nmap
3) DIRB
4) Exit
Enter your choice [1-4]: 3
[+] Running DIRB against 10.15.203.91

-----
DIRB v2.22
```

Figure 9 — DIRB starting up after being selected from the menu.

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

DIRB v2.22
By The Dark Raver

START_TIME: Sat Sep 5 22:53:35 2026
URL_BASE: http://10.15.203.91/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

GENERATED WORDS: 4612

— Scanning URL: http://10.15.203.91/ —
+ http://10.15.203.91/cgi-bin/ (CODE:403|SIZE:293)
⇒ DIRECTORY: http://10.15.203.91/dav/
```

Figure 10 — DIRB successfully discovering directories, including phpMyAdmin, on the target.

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

Home
⇒ DIRECTORY: http://10.15.203.91/dav/
+ http://10.15.203.91/index (CODE:200|SIZE:891)
+ http://10.15.203.91/index.php (CODE:200|SIZE:891)
+ http://10.15.203.91/phpinfo (CODE:200|SIZE:48062)
+ http://10.15.203.91/phpinfo.php (CODE:200|SIZE:48074)

Trash
⇒ DIRECTORY: http://10.15.203.91/phpMyAdmin/
+ http://10.15.203.91/server-status (CODE:403|SIZE:298)

⇒ DIRECTORY: http://10.15.203.91/test/

⇒ DIRECTORY: http://10.15.203.91/twiki/

— Entering directory: http://10.15.203.91/dav/ —

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

Home
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/phpMyAdmin/ —

File System
+ http://10.15.203.91/phpMyAdmin/calendar (CODE:200|SIZE:4145)
+ http://10.15.203.91/phpMyAdmin/changelog (CODE:200|SIZE:74593)
+ http://10.15.203.91/phpMyAdmin/ChangeLog (CODE:200|SIZE:40540)

⇒ DIRECTORY: http://10.15.203.91/phpMyAdmin/contrib/
+ http://10.15.203.91/phpMyAdmin/docs (CODE:200|SIZE:4583)
+ http://10.15.203.91/phpMyAdmin/error (CODE:200|SIZE:1063)
+ http://10.15.203.91/phpMyAdmin/export (CODE:200|SIZE:4145)
+ http://10.15.203.91/phpMyAdmin/favicon.ico (CODE:200|SIZE:18902)
+ http://10.15.203.91/phpMyAdmin/import (CODE:200|SIZE:4145)
+ http://10.15.203.91/phpMyAdmin/index (CODE:200|SIZE:4145)
+ http://10.15.203.91/phpMyAdmin/index.php (CODE:200|SIZE:4145)

⇒ DIRECTORY: http://10.15.203.91/phpMyAdmin/js/
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
=> DIRECTORY: http://10.15.203.91/phpMyAdmin/js/

=> DIRECTORY: http://10.15.203.91/phpMyAdmin/lang/

=> DIRECTORY: http://10.15.203.91/phpMyAdmin/libraries/
+ http://10.15.203.91/phpMyAdmin/license (CODE:200|SIZE:18011)
+ http://10.15.203.91/phpMyAdmin/LICENSE (CODE:200|SIZE:18011)
+ http://10.15.203.91/phpMyAdmin/main (CODE:200|SIZE:4227)
+ http://10.15.203.91/phpMyAdmin/navigation (CODE:200|SIZE:4145)
+ http://10.15.203.91/phpMyAdmin/phpinfo (CODE:200|SIZE:0)
+ http://10.15.203.91/phpMyAdmin/phpinfo.php (CODE:200|SIZE:0)
+ http://10.15.203.91/phpMyAdmin/phpmyadmin (CODE:200|SIZE:21389)
+ http://10.15.203.91/phpMyAdmin/print (CODE:200|SIZE:1063)
+ http://10.15.203.91/phpMyAdmin/readme (CODE:200|SIZE:2624)
+ http://10.15.203.91/phpMyAdmin/README (CODE:200|SIZE:2624)
+ http://10.15.203.91/phpMyAdmin/robots (CODE:200|SIZE:26)
+ http://10.15.203.91/phpMyAdmin/robots.txt (CODE:200|SIZE:26)

=> DIRECTORY: http://10.15.203.91/phpMyAdmin/scripts/
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help
+ http://10.15.203.91/phpMyAdmin/README (CODE:200|SIZE:2624)
+ http://10.15.203.91/phpMyAdmin/robots (CODE:200|SIZE:26)
+ http://10.15.203.91/phpMyAdmin/robots.txt (CODE:200|SIZE:26)

=> DIRECTORY: http://10.15.203.91/phpMyAdmin/scripts/

=> DIRECTORY: http://10.15.203.91/phpMyAdmin/setup/
+ http://10.15.203.91/phpMyAdmin/sql (CODE:200|SIZE:4145)

=> DIRECTORY: http://10.15.203.91/phpMyAdmin/test/

=> DIRECTORY: http://10.15.203.91/phpMyAdmin/themes/
+ http://10.15.203.91/phpMyAdmin/TOD0 (CODE:200|SIZE:235)
+ http://10.15.203.91/phpMyAdmin/webapp (CODE:200|SIZE:6900)

— Entering directory: http://10.15.203.91/test/ —

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/twiki/ —

⇒ DIRECTORY: http://10.15.203.91/twiki/bin/
+ http://10.15.203.91/twiki/data (CODE:403|SIZE:295)
+ http://10.15.203.91/twiki/index (CODE:200|SIZE:782)
+ http://10.15.203.91/twiki/index.html (CODE:200|SIZE:782)

⇒ DIRECTORY: http://10.15.203.91/twiki/lib/
+ http://10.15.203.91/twiki/license (CODE:200|SIZE:19440)

⇒ DIRECTORY: http://10.15.203.91/twiki/pub/
+ http://10.15.203.91/twiki/readme (CODE:200|SIZE:4334)
+ http://10.15.203.91/twiki/templates (CODE:403|SIZE:300)
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

+ http://10.15.203.91/twiki/readme (CODE:200|SIZE:4334)
+ http://10.15.203.91/twiki/templates (CODE:403|SIZE:300)

— Entering directory: http://10.15.203.91/phpMyAdmin/contrib/ --
--

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/phpMyAdmin/js/ —

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/phpMyAdmin/lang/ —

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/phpMyAdmin/scripts/ --
--
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/phpMyAdmin/setup/ —
+ http://10.15.203.91/phpMyAdmin/setup/config (CODE:303|SIZE:1370)

⇒ DIRECTORY: http://10.15.203.91/phpMyAdmin/setup/frames/
+ http://10.15.203.91/phpMyAdmin/setup/index (CODE:200|SIZE:8616)
+ http://10.15.203.91/phpMyAdmin/setup/index.php (CODE:200|SIZE:8624)
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

+ http://10.15.203.91/phpMyAdmin/setup/index.php (CODE:200|SIZE:8624)

⇒ DIRECTORY: http://10.15.203.91/phpMyAdmin/setup/lib/
+ http://10.15.203.91/phpMyAdmin/setup/scripts (CODE:200|SIZE:21967)
+ http://10.15.203.91/phpMyAdmin/setup/styles (CODE:200|SIZE:6218)

— Entering directory: http://10.15.203.91/phpMyAdmin/test/ —

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/phpMyAdmin/themes/ —
-

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

- Home
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/twiki/bin/ —

Trash
(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/twiki/lib/ —

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

— Entering directory: http://10.15.203.91/twiki/pub/ —

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)
```

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

— Entering directory: http://10.15.203.91/phpMyAdmin/setup/frame
s/ —

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

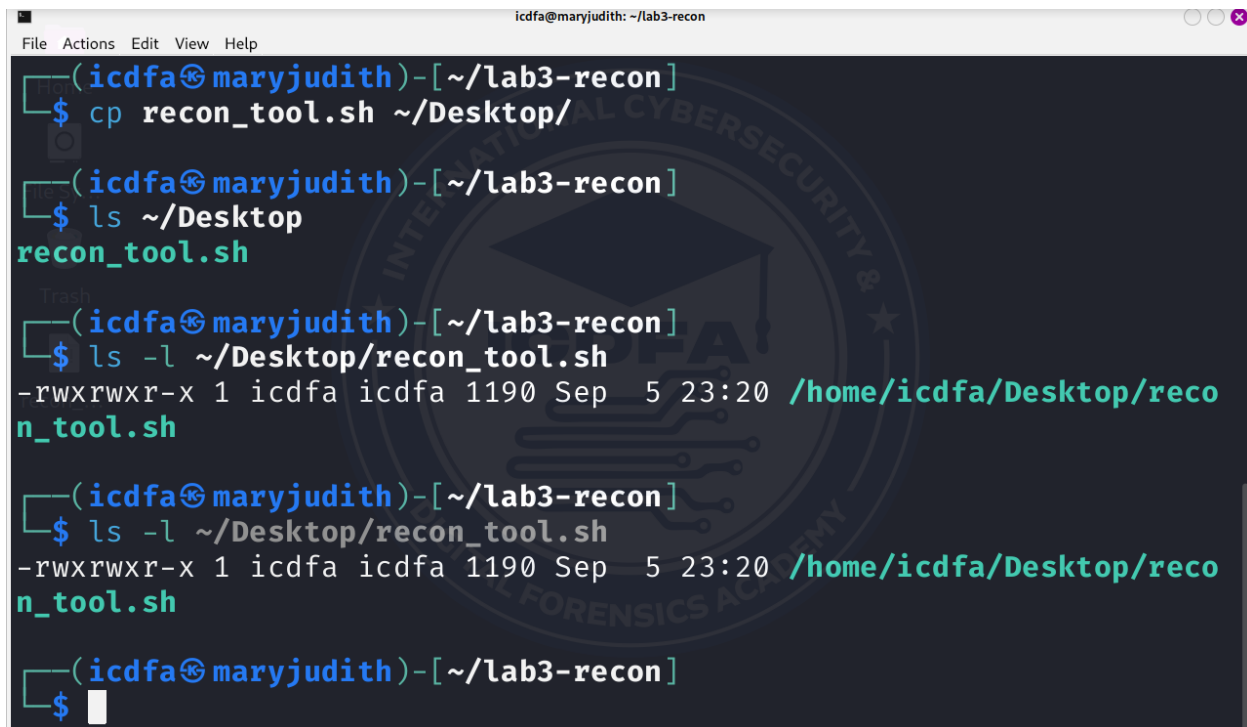
— Entering directory: http://10.15.203.91/phpMyAdmin/setup/lib/
—

(!) WARNING: Directory IS LISTABLE. No need to scan it.
(Use mode '-w' if you want to scan it anyway)

END_TIME: Sat Sep 5 22:54:13 2026
DOWNLOADED: 18448 - FOUND: 42

(icdfa@maryjudith)-[~/lab3-recon]
$
```

With **WhatWeb**, **Nmap** and **DIRB** all completing successfully against the same authorized target, the assessment's requirement of "at least two tools" executing successfully was comfortably met

A terminal window titled 'icdfa@maryjudith: ~/lab3-recon' with a menu bar (File, Actions, Edit, View, Help). The terminal shows a sequence of commands and their outputs. First, 'cp recon_tool.sh ~/Desktop/' is executed. Then, 'ls ~/Desktop' is run, showing 'recon_tool.sh'. Next, 'ls -l ~/Desktop/recon_tool.sh' is run, displaying file permissions '-rwxrwxr-x', owner 'icdfa', group 'icdfa', size '1190', date 'Sep 5 23:20', and path '/home/icdfa/Desktop/recon_tool.sh'. This last command is repeated. The terminal background features a faint watermark of a graduation cap and the text 'INTERNATIONAL CYBERSECURITY & DIGITAL FORENSICS ACADEMY'.

```
icdfa@maryjudith: ~/lab3-recon
File Actions Edit View Help

(icdfa@maryjudith)-[~/lab3-recon]
$ cp recon_tool.sh ~/Desktop/

(icdfa@maryjudith)-[~/lab3-recon]
$ ls ~/Desktop
recon_tool.sh

Trash
(icdfa@maryjudith)-[~/lab3-recon]
$ ls -l ~/Desktop/recon_tool.sh
-rwxrwxr-x 1 icdfa icdfa 1190 Sep  5 23:20 /home/icdfa/Desktop/reco
n_tool.sh

(icdfa@maryjudith)-[~/lab3-recon]
$ ls -l ~/Desktop/recon_tool.sh
-rwxrwxr-x 1 icdfa icdfa 1190 Sep  5 23:20 /home/icdfa/Desktop/reco
n_tool.sh

(icdfa@maryjudith)-[~/lab3-recon]
$
```

Figure 11: Script Backup and Verification

The recon_tool.sh file was successfully copied to the Desktop and its presence was verified using the ls command. File permissions were also confirmed using ls -l, showing that the script remained executable and ready for submission.

Challenges Encountered

During the completion of this lab, I encountered the following challenges:

- I mistakenly typed `ls l` instead of `ls -l`, which resulted in an error.
- Care was required when handling quotation marks around `"$target"` to avoid Bash syntax errors.
- Building the script incrementally proved more effective than attempting to create the entire program at once.

What I Learned

From this lab, I learned:

- How to collect user input using `read -rp`.
- How to validate input using the `-z` operator.
- How to use functions in Bash.
- How to create menu-driven programs using the case statement.
- How to verify program availability using command `-v`.
- How to grant execute permissions using `chmod +x`.
- How to automate reconnaissance activities using Bash scripting.
- The importance of conducting reconnaissance only against authorized systems.

Answers to Understanding Questions

1. What does the shebang do?

It tells Linux which interpreter should execute the script.

2. What does `read -rp` do?

It displays a prompt and stores user input in a variable.

3. Why quote `"$target"`?

To safely handle spaces and special characters.

4. What does `-z` test?

It checks whether a string is empty.

5. What is the purpose of `exit 1`?

It terminates the script and indicates an error condition.

6. Why is case suitable for menus?

It provides a clean and readable way to manage multiple options.

7. What does;; mean?

It marks the end of a branch within a case statement.

8. What does * mean in a case statement?

It acts as the default or catch-all branch.

9. Why do WhatWeb and DIRB use http://\$target?

Because both tools work against web servers and require a URL format.

10. What does nmap -sV do?

It detects service versions running on open ports.

11. What does command -v check?

It checks whether a command exists in the system PATH.

12. What does chmod +x do?

It grants execute permission to a file.

13. What does ./ mean?

It tells the shell to run a file from the current directory.

14. What does tee do?

It displays command output and saves it to a file simultaneously.

15. Which requirement prevents hardcoding?

The requirement that the target must be supplied by the user.

Conclusion

This lab successfully demonstrated how Bash scripting can be used to automate reconnaissance activities through a simple menu-driven tool. The completed script prompts the user for a target, validates input, checks for required dependencies, and launches WhatWeb, Nmap, or DIRB based on the selected option. Testing against the authorized target **10.15.203.91** confirmed successful execution of all three reconnaissance tools. The exercise strengthened my understanding of Bash scripting concepts such as variables, functions, input validation, conditional statements, and case logic, while also reinforcing the importance of authorization and ethical conduct during security assessments.